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Emotion Recognition in Autistic Children Through Facial Expressions Using Advanced Deep Learning Architectures

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Abstract

Atypical and subtle facial expression patterns in individuals with autism spectrum disorder (ASD) pose a significant challenge for automated emotion recognition. This study evaluates and compares the performance of convolutional neural networks (CNNs) and transformer-based deep learning models for facial emotion recognition in this population. Using a labeled dataset of emotional facial images, we assessed eight models across four emotion categories: natural, anger, fear, and joy. Our results demonstrate that transformer models consistently outperformed CNNs in both overall and emotion-specific metrics. Notably, the Swin Transformer achieved the highest performance, with an accuracy of 0.8000 and an F1-score of 0.7889, significantly surpassing all CNN counterparts. While CNNs failed to detect the fear class, transformer models showed a measurable capability in identifying complex emotions such as anger and fear, suggesting an enhanced ability to capture subtle facial cues. Analysis of the confusion matrix further confirmed the transformers' superior classification balance and generalization. Despite these promising results, the study has limitations, including class imbalance and its reliance solely on facial imagery. Future work should explore multimodal emotion recognition, model interpretability, and personalization for real-world applications. Research also demonstrates the potential of transformer architectures in advancing inclusive, emotion-aware AI systems tailored for autistic individuals.

Keywords: emotion recognition; autism spectrum disorder; transformer model; deep learning; Swin Transformer; affective computing

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1. Introduction

Autism spectrum disorder (ASD) is a complex neurodevelopmental condition that affects approximately 1 in 54 children and is marked by persistent challenges in social communication, emotional understanding, and behavioral regulation [1,2]. While the manifestations of ASD vary widely among individuals, a hallmark characteristic across the spectrum is difficulty with recognizing and interpreting facial expressions—skills that are critical for establishing and maintaining social relationships [3]. The inability to accurately perceive and respond to emotional cues in others can lead to breakdowns in communication, limited peer interactions, and significant social isolation over time [4,5]. Understanding and enhancing emotion recognition in autistic children is therefore of paramount importance for improving their quality of life and supporting their developmental outcomes [3,6]. Research has shown that autistic individuals tend to exhibit atypical gaze patterns when

viewing human faces [7]. Unlike neurotypical individuals, who instinctively focus on emotionally salient regions of the face such as the eyes and mouth, autistic individuals may direct their attention to less informative areas or distribute their gaze in ways that miss key social cues [8]. This divergence in facial processing has been linked to measurable deficits in emotional interpretation and is believed to contribute to the broader social difficulties experienced by individuals with ASD. Consequently, accurate assessment and training in facial emotion recognition are becoming a foundational element of therapeutic interventions [9].

Traditionally, assessments of emotional recognition in ASD populations have relied heavily on behavioral observations, structured interviews, and standardized diagnostic questionnaires administered by clinicians or caregivers [10,11]. While these tools offer valuable insights, they suffer from several limitations: they are time-consuming to administer, highly dependent on the observer's interpretation, and often lack the granularity needed to capture subtle emotional processing differences. Furthermore, these assessments are not easily scalable for large populations or routine screening in clinical or educational settings [11]. The variability in expressive behaviors among individuals with ASD further complicates consistent evaluation, underscoring the need for more objective, efficient, and scalable assessment methods.

Recent advancements in artificial intelligence and deep learning have opened new frontiers for emotion recognition, particularly through the use of image-based techniques. Deep learning algorithms, and in particular CNNs, have demonstrated exceptional performance in visual classification tasks, including facial expression recognition [5,6]. These models are capable of automatically learning hierarchical feature representations from raw pixel data, making them ideally suited for tasks that require fine-grained interpretation of facial expressions [12–14]. Unlike traditional machine learning methods, which require extensive manual feature engineering, CNNs can identify relevant facial structures and emotional indicators without human intervention, thus increasing accuracy while reducing bias and subjectivity [3].

In addition to CNNs, transformer-based architectures such as Vision Transformers (ViTs) have emerged as powerful tools in the domain of image analysis. ViTs utilize self-attention mechanisms to capture long-range dependencies and global contextual relationships across the entire image, providing complementary advantages to the localized pattern recognition capabilities of CNNs [15]. In the context of emotion recognition, ViTs may offer enhanced sensitivity to subtle, distributed changes in facial expressions that are not easily captured by spatially localized filters alone [16]. The usage of these two approaches—CNNs for detailed, localized feature extraction and ViTs for capturing broader contextual patterns—represents a promising direction for emotion recognition in complex, real-world scenarios.

Despite their demonstrated success in general-purpose emotion recognition tasks, the application of deep learning models to ASD-specific datasets remains underdeveloped. Most existing systems are trained on data collected from neurotypical populations, which may not adequately represent the facial expressiveness or emotional display patterns of autistic individuals [17,18]. This misalignment can result in biased models that perform poorly when deployed in real-world settings involving autistic users. The heterogeneity of ASD further complicates model training, as individuals may differ significantly in how they display emotions—both in terms of facial muscle activation and temporal dynamics [19]. Consequently, there is a critical need to develop models that are robust to this variability and capable of generalizing across the spectrum of expressive behaviors seen in autism. Another major barrier to progress in this area is the lack of representative training data. Many widely used facial expression datasets, such as FER2013 or AffectNet, contain limited examples

of autistic individuals and often focus only on exaggerated, prototypical emotions [20,21]. This lack of inclusivity limits the generalizability of models trained on these datasets. To overcome this limitation, recent research has begun to explore the use of ASD-specific datasets that include images or videos of autistic children displaying a range of emotional expressions [22–24]. Such datasets enable the development of models that are better attuned to the subtle and diverse ways emotions are expressed within the ASD population.

In this study, a deep learning-based approach has been employed to improve the classification of facial emotions in autistic children. Recognizing the essential role of emotion interpretation in social development, the study utilizes state-of-the-art deep learning techniques, including CNNs and transfer learning, to build specialized models for emotion recognition. By leveraging pre-trained architectures—such as ResNet, DenseNet, or Inception networks—and fine-tuning them on autism-relevant datasets, the models have been adapted for multi-class classification tasks involving four fundamental emotions: natural, anger, fear, and joy [25]. These emotions were selected based on their social relevance and frequency of occurrence in early childhood interactions. The use of transfer learning significantly accelerates the training process and improves performance, especially when dealing with limited or imbalanced datasets [26]. Pre-trained models, originally trained on large-scale image databases, retain a broad understanding of general visual features that can be refined to specialize in emotion recognition. Fine-tuning these models on ASD-specific data helps to bridge the gap between general vision tasks and the domain-specific challenges of emotion classification in autistic children. By optimizing the performance of these models through careful selection of hyperparameters, data augmentation techniques, and model architecture adjustments, this study demonstrates the practical viability of deploying deep learning for emotion recognition in autism contexts. The findings support the notion that transfer learning not only improves classification accuracy but also enhances the model's sensitivity to less pronounced emotional cues that are common in ASD populations.

The implementation of such systems holds substantial promise for real-world applications. Emotion-aware tools can be integrated into educational software, therapeutic games, and communication aids, providing real-time feedback and support to children during social interactions [6,11]. These tools can also assist therapists and educators by offering quantitative, objective metrics for monitoring emotional understanding and social engagement over time [4]. In addition, mobile or tablet-based implementations allow for widespread deployment in schools, clinics, or home environments—especially in regions where access to specialized care may be limited [27].

Despite progress in ASD-related emotion recognition using deep learning, several research gaps remain. First, many existing emotion classifiers are not designed with the variability of autistic expression in mind and may underperform when tested on diverse ASD populations [21,28]. Second, while deep learning architectures such as CNNs and transformers offer powerful modeling capabilities, there has been limited exploration into how different architectures—individually or in combination—perform specifically in autism-related contexts [17]. Understanding the trade-offs between spatial and contextual feature extraction remains an open question. Third, most emotion recognition systems are limited to a narrow set of basic emotions. Future research should examine the recognition of more nuanced or compound emotions, as well as the temporal dynamics of facial expressions [10,28]. The unfolding of emotion over time may provide critical cues, particularly in autism, where expression timing and response delays can differ from neurotypical patterns. Finally, computational efficiency is an ongoing concern. To enhance accessibility and real-world applicability, emotion recognition models must be optimized for deployment on low-power, resource-constrained devices without significant loss of accuracy [29].

This research contributes to the growing field of autism-related emotion recognition and deep learning in several key ways:

- It compares CNN and transformer-based architectures for emotion classification in autistic children, evaluating their strengths and limitations in capturing subtle facial cues.
- It assesses the role of dataset composition and transfer learning in optimizing model accuracy and generalizability across diverse ASD expressions.
- It demonstrates the potential for scalable, real-time, and objective emotion-aware systems that can augment therapeutic interventions, social skills training, and assistive communication technologies.

The remainder of this paper is organized as follows: Section 2 reviews relevant work in deep learning, autism research, and emotion recognition. Section 3 describes the methodology, including dataset selection, model architectures, and training protocols. Section 4 presents the experimental results and model comparisons. Section 5 concludes with key findings and directions for future work.

2. Related Works

Recent advancements in deep learning have considerably improved the ability of automated systems to analyze human behavior, including complex tasks such as emotion recognition. Within the context of ASD, the accurate and timely identification of emotions through facial expressions is critical for interpreting the distinctive communication styles of autistic individuals and informing personalized interventions. Despite progress in this area, the development of deep learning models that are both accurate and generalizable—while remaining practical for deployment—remains an ongoing research challenge. This is particularly true given the atypical and often subtle nature of emotional expression in autistic children.

A substantial portion of the existing literature has explored the application of deep learning techniques for the diagnosis and early screening of ASD using facial image analysis. For example, Li et al. [30] introduced a two-stage transfer learning framework incorporating MobileNetV2 and MobileNetV3-Large, achieving an area under the curve (AUC) of 96.32% and an accuracy of 90.5% on ASD screening tasks, with optimizations tailored for mobile deployment. Ranjana and Muthukkumar [31] proposed a DenseResNet-based architecture that combines DenseNet and ResNet models, achieving a classification accuracy of 97.07% on a large dataset. Similarly, Akter et al. [32] employed MobileNetV1 within a transfer learning paradigm, reporting 90.67% accuracy for ASD classification. Although these contributions demonstrate the effectiveness of deep learning in extracting ASD-related facial features, their primary focus is on detecting ASD presence rather than on the more granular task of emotional state recognition. This distinction is significant, as the methodologies employed in ASD diagnosis may not directly translate to the domain of emotion recognition, where subtler distinctions are required. More directly relevant to emotion analysis, several studies have focused on the recognition of facially expressed emotional or affective states—including pain—in autistic children. Talaat et al. [28] presented a real-time system for emotion identification that integrates a deep convolutional neural network (DCNN) with a kernel autoencoder for feature extraction. Their system demonstrated strong performance, particularly when employing the Xception model, achieving an accuracy of 95.23% across six emotional categories. The integration of fog and Internet of Things (IoT) technologies further supported real-time implementation. Afrin et al. [33] developed a hybrid architecture that fuses DenseNet121 and MobileNetV2 to identify four emotional states (joy, anger, natural, and fear) in autistic children. Notably, they addressed significant limitations in existing datasets—such as class overlap and image

duplication—by curating a modified dataset, FERAC (Facial Emotion Recognition-Autistic Children), consisting of 770 images. This dataset was constructed through observational data collection under expert supervision, leading to the exclusion of problematic classes like 'Sadness' and 'Surprise'. The hybrid model trained on FERAC achieved an accuracy of 75%, outperforming several baseline architectures; however, the overall performance still indicates room for enhancement in generalizability and robustness. In a related line of work, Sandeep and Kumar [34] proposed a system for pain recognition in autistic children by combining ResNeXt and MediaPipe with a convolutional neural network to assess facial indicators of pain in real time. Although highly specific in scope, this study exemplifies the potential of deep learning for recognizing complex affective states in ASD populations. Multimodal frameworks have also been investigated to enhance emotion recognition performance by integrating visual and auditory features. Wang et al. [35], for instance, proposed a Convolutional Vision Transformer model that incorporates both facial image data and speech-derived features. Their system achieved facial-only emotion recognition accuracy of 79.12%, which improved to 90.73% with multimodal feature fusion using an attention mechanism. While such approaches yield higher performance, they also introduce considerable computational overhead and practical limitations related to real-time data synchronization, particularly in settings where visual cues are the primary or sole modality of interest. Moreover, the suboptimal performance of unimodal facial emotion recognition reported in such studies indicates that further advancements are needed to improve the efficacy of visual-only models.

Collectively, these works highlight several ongoing challenges and research gaps. First, a majority of studies in this field emphasize ASD classification rather than comprehensive emotion recognition. Among those that do address emotional states, there is often a lack of consistent accuracy and model robustness when applied to the inherently diverse and sometimes ambiguous facial expressions observed in autistic individuals. The quality, diversity, and class balance of training datasets also remain critical bottlenecks, as unbalanced or insufficiently representative datasets hinder the development of generalized models. Furthermore, the computational complexity of many current models—especially those employing multimodal or ensemble methods—poses constraints on their applicability in clinical or real-time environments, where efficiency and low-latency performance are essential. The atypical nature of affective expression in autistic children further necessitates model architectures that are finely attuned to detecting subtle and non-standard emotional cues.

In response to these challenges, the present study proposes a deep learning-based approach specifically designed for emotion recognition in autistic children using static facial images. The goal is to develop models that are both accurate and computationally efficient, capable of recognizing a diverse array of emotions while being suitable for deployment in real-world contexts. By addressing dataset limitations, refining model architectures, and targeting a broader emotional spectrum, this work aims to advance the state of emotion recognition within ASD-focused research and applications.

3. Materials and Methods

We evaluated deep learning approaches for emotion recognition in individuals with ASD, focusing on the comparative performance of CNNs and transformer-based architectures. Our workflow consists of three main steps, as shown in Figure 1: (1) preprocessing facial expression data into four target emotion classes—natural, anger, fear, and joy—based on a curated ASD-specific dataset; (2) training and validation CNN and transformer models independently to assess their performance in emotion classification; and (3) evaluating classification accuracy and standardized metrics.

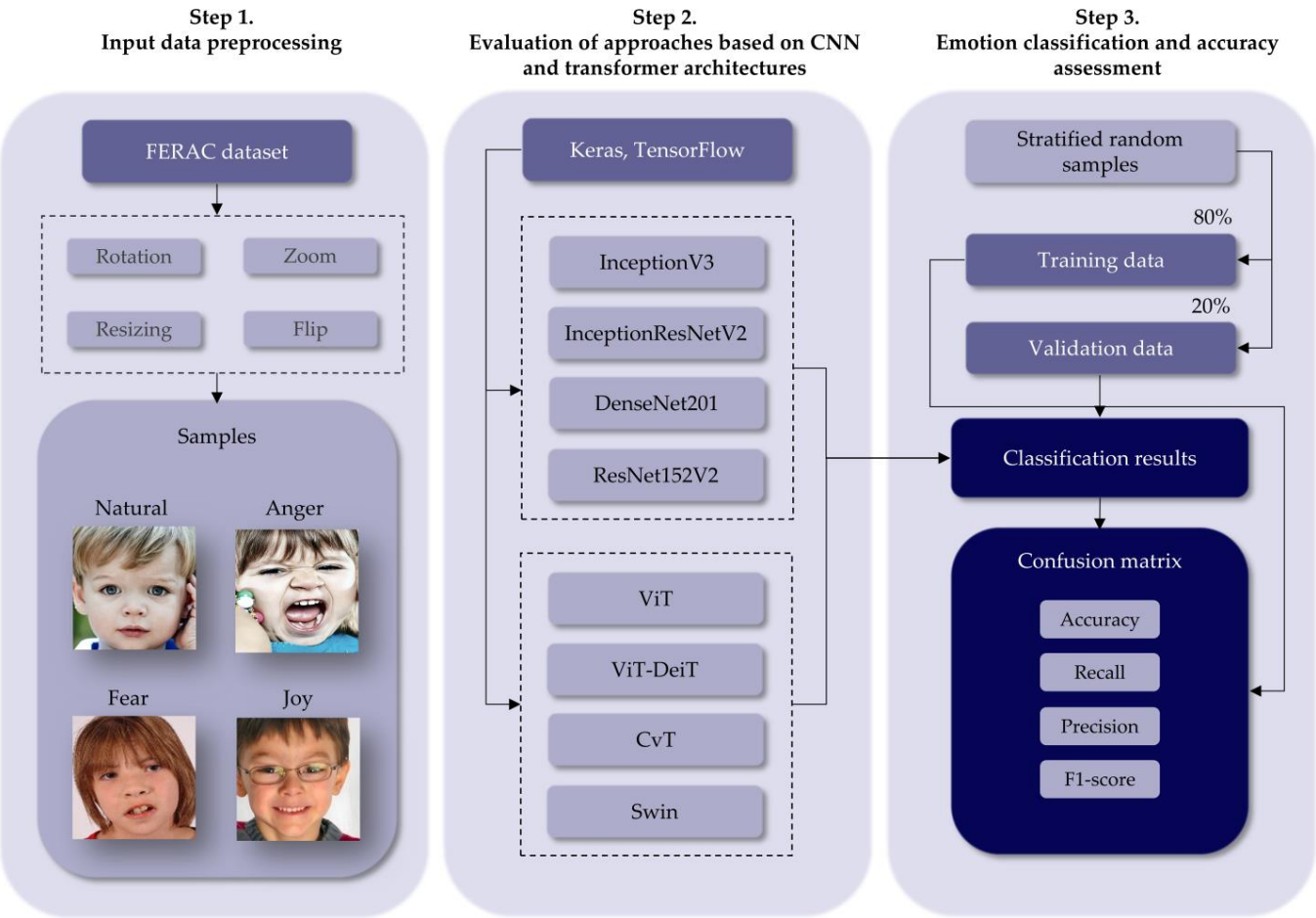

Figure 1. Workflow for emotion recognition in children with autism using deep learning. Step 1 involves preprocessing FERAC facial expression data with augmentation, including rotation, zoom, resizing, and flipping across four emotion classes: natural, anger, fear, and joy. Step 2 consists of training and evaluating CNN models (InceptionV3, InceptionResNetV2, DenseNet201, ResNet152V2) and transformer-based models (ViT, ViT-DeiT, CvT, Swin) using Keras and TensorFlow. Step 3 covers emotion classification and performance evaluation using an 80/20 stratified train-validation split with metrics such as accuracy, precision, recall, and F1-score derived from the confusion matrix.

3.1. Data Preprocessing and Experimental Setup

We utilized the Facial Emotion Recognition-Autistic Children (FERAC) Dataset, which consists of 770 images of children's faces [25]. This dataset was developed by modifying an existing one, in collaboration with a medical professional who is the director of the Autism Development Centre at Ma Shishu O General Hospital, Chattogram, Bangladesh. Initially, all black and white and duplicate images were removed to ensure high data quality. The dataset comprises four emotional classes: natural, fear, joy, and anger [25]. For our experimental setup, we partitioned the dataset into 691 images for the training directory and 79 images for the testing directory. Figure 2 shows sample images, while Table 1 displays the dataset's distribution.

Figure 2. Samples of emotional facial expression images representing different emotions were used in the experiment.

Table 1. Dataset distribution.

Category	Total Images	Training Images	Validation Images
Natural	184	147	37
Anger	74	59	15
Fear	49	39	10
Joy	463	370	93
Total	770	615	155

For model input consistency, all images were first resized to 224 × 224 pixels. The deep learning models were then implemented and trained in a Python 3.10 (Python Software Foundation, Wilmington, DE, USA) environment on Kaggle. We utilized the Keras-GPU and TensorFlow-GPU frameworks for this task. Training was performed using an NVIDIA Tesla P100 GPU for enhanced speed. The training protocol consisted of 10 epochs with a batch size of 16. The Adam optimizer was employed for dynamic updates, and an adaptive learning rate mechanism was also included to prevent overfitting and facilitate better model convergence.

3.2. CNN and Transfer Learning for Emotion Classification

CNNs have become the foundation of most modern computer vision applications due to their ability to detect hierarchical patterns in image data [36]. For emotion recognition, CNNs are effective in identifying local features such as eye movement, mouth shape, and facial muscle patterns [37,38].

To enhance the generalization capability and reduce the training time, transfer learning was adopted. Instead of training a CNN from scratch, pre-trained models such as ResNet or DenseNet, originally trained on large-scale image datasets like ImageNet, were used as a base [32]. These models were then fine-tuned on the emotion dataset to adapt the learned features to the specific task of emotion recognition in children.

The training process involved standard preprocessing steps, including resizing, normalization, and data augmentation. Augmentation techniques such as rotation, flipping, and scaling were applied to improve model robustness and mitigate overfitting. The final layers of the CNN were customized to include dense layers and a softmax classifier suited to the multi-class nature of the task. The whole process is shown in Algorithm 1.

Algorithm 1: CNN with Transfer Learning

```
function EmotionRecognition_CNN (Input_Images, Pretrained_Model)
    Input:
        Input_Images: Set of training images with emotion labels
        Pretrained_Model: A CNN model pre-trained on ImageNet
    Output: Trained_CNN_Model
    Preprocess Input_Images (resize, normalize, augment)
    Load Pretrained_Model without top classification layers
    Freeze early layers of Pretrained_Model to retain learned features
    Add custom classification layers:
        Dense layer with ReLU activation
        Dropout for regularization
        Final Dense layer with Softmax activation for emotion classes
    Compile model using Adam optimizer and cross-entropy loss
    Train model on Input_Images with validation split
    Evaluate model performance on test set
    return Trained_CNN_Model
end function
```

3.3. Transformer-Based Approach for Emotion Classification

Transformers, originally designed for natural language processing tasks, have recently shown impressive performance in computer vision problems [39]. Vision Transformers (ViTs) process images by dividing them into smaller patches and interpreting these patches as a sequence of tokens, similar to how words are processed in sentences [40]. This allows the model to capture global contextual relationships that CNNs might miss.

In this study, a pre-trained transformer model was used. They are specialized Vision Transformers fine-tuned for facial emotion recognition. The model processes high-resolution image inputs and uses multi-head self-attention mechanisms to capture complex emotional patterns distributed across different facial regions.

The feature extraction stage involved passing preprocessed images through the frozen encoder to generate deep feature representations. These features were then used to train a custom classifier composed of dense layers, as shown in Algorithm 2. This two-stage approach allowed for leveraging powerful representations while reducing computational cost during training.

Algorithm 2: Transformer-Based Emotion Classification

```
function EmotionRecognition_Transformer(Input_Images, Transformer_Model)
    Input:
        Input_Images: Set of facial images labeled with emotions
        Transformer_Model: Pretrained model for emotion analysis
    Output: Trained_Transformer_Model
    Preprocess Input_Images:
        Convert grayscale to RGB if necessary
        Normalize pixel values and apply standard augmentation
    Use Transformer_Model to extract deep features from Input_Images
    Store extracted features and corresponding labels
    Define custom classifier head:
        Dense layer with ReLU activation
        Dropout for regularization
        Final Dense layer with Softmax activation
    Compile model with suitable optimizer and loss function
    Train classifier head using extracted features and labels
    Evaluate model using accuracy, F1-score, and AUC
    return Trained_Transformer_Model
end function
```

3.4. Performance Assessment

We evaluated the performance of our deep learning models for multi-class classification across four distinct classes. We analyzed the training and validation accuracy, loss, and key classification metrics derived from the confusion matrix. These metrics—precision, recall, F1-score, and accuracy, as defined in Equations (1)–(4)—allowed us to perform a comprehensive evaluation of the models' capability across all classes.

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \tag{1}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \tag{2}$$

$$\text{F1-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}, \tag{3}$$

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FP} + \text{TN} + \text{FN}}, \tag{4}$$

4. Results and Discussion

In terms of overall model performance, transformer-based architectures consistently outperformed CNN-based models across all major evaluation metrics, as shown in Table 2.

Among the CNNs, ResNet152V2 achieved the highest accuracy at 0.7484, along with the best F1-score of 0.7048 and precision of 0.7328 within this category. The next best-performing CNN was DenseNet201, which reached an accuracy of 0.7355 and an F1-score of 0.7007, showing a modest trade-off between recall of 0.7355 and precision of 0.6755. InceptionV3 and InceptionResNetV2 showed slightly lower performance, with InceptionV3 achieving an accuracy of 0.7290 and an F1-score of 0.6868, while InceptionResNetV2 trailed with 0.7161 accuracy and a 0.6774 F1-score.

Table 2. Detailed performance metrics of CNN and transformer models on emotion recognition in autistic children.

Deep Learning Approach	Model	Accuracy	F1-Score	Precision	Recall
CNN	InceptionV3	0.7290	0.6868	0.6802	0.7290
	InceptionResNetV2	0.7161	0.6774	0.6532	0.7161
	DenseNet201	0.7355	0.7007	0.6755	0.7355
	ResNet152V2	**0.7484**	**0.7048**	**0.7328**	**0.7484**
Transformer	ViT	0.7613	0.7341	0.7272	0.7613
	ViT-DeiT	0.7484	0.7395	0.7347	0.7484
	CvT	0.7613	0.7300	0.7525	0.7613
	Swin	**0.8000**	**0.7889**	**0.7903**	**0.8000**

The highest assessment metrics are bolded.

In contrast, transformer models exhibited notably stronger performance. The Swin Transformer outperformed all other models, achieving the highest accuracy of 0.8000, an F1-score of 0.7889, precision of 0.7903, and recall of 0.8000. These results suggest that Swin not only excels in correctly identifying the target classes but also maintains a strong balance between sensitivity and specificity. ViT and CvT also performed well, both reaching an accuracy of 0.7613. While ViT demonstrated a higher F1-score of 0.7341, CvT had the advantage in precision of 0.7525. ViT-DeiT, a distilled variant of the Vision Transformer, matched ResNet152V2 in accuracy of 0.7484 but exceeded it in F1-score of 0.7395, indicating more balanced predictions.

These findings highlight the relative superiority of transformer-based models over traditional CNNs in the context of emotion recognition in autistic children, especially when considering the overall balance between the evaluated metrics.

Further analysis of emotion-specific F1-scores provides insight into how each model performs across the individual emotion categories: natural, anger, fear, and joy, as shown in Table 3.

Table 3. The F1-score values for emotion recognition in autistic children per model.

Deep Learning Approach	Model	Natural	Anger	Fear	Joy
CNN	InceptionV3	0.5676	0.4000	0.0000	0.8544
	InceptionResNetV2	0.5352	**0.4167**	0.0000	0.8488
	DenseNet201	0.5882	0.4138	0.0000	0.8670
	ResNet152V2	**0.6341**	0.2353	0.0000	**0.8844**
Transformer	ViT	0.5312	0.4444	0.2667	0.9118
	ViT-DeiT	0.5075	0.6000	0.3000	0.9016
	CvT	0.6076	0.1667	0.1818	**0.9286**
	Swin	**0.6269**	**0.6400**	**0.4000**	0.9192

The highest assessment metrics are bolded.

For the joy emotion, which had the highest overall recognition rates across models, transformer architectures again led the performance. CvT achieved the best result with an F1-score of 0.9286, followed closely by Swin (0.9192), ViT (0.9118), and ViT-DeiT (0.9016). Among CNNs, the top performer was ResNet152V2 with an F1-score of 0.8844. This suggests that joy-related facial expressions are relatively easier to classify, particularly for transformer-based models. The fear category posed the greatest challenge for all models, with CNNs failing to recognize this emotion entirely with an F1-score of 0.0000. In contrast, transformer models showed some capability, albeit limited. Swin achieved the highest F1-score at 0.4000, followed by ViT-DeiT (0.3000), ViT (0.2667), and CvT (0.1818). While the performance remains suboptimal, the transformers' ability to detect fearful expressions, even to a small extent, is notable and may be a starting point for further refinement. Recognition of anger was moderate across models, with transformer models again demonstrating superiority. Swin reached the highest F1-score of 0.6400, followed by ViT-DeiT (0.6000) and ViT (0.4444). CNNs performed relatively poorly, with scores ranging from 0.2353 for ResNet152V2 to 0.4167 for InceptionResNetV2. These results suggest that anger expressions are more difficult to recognize than joy but easier than fear, with transformer models showing a clear advantage. In the natural category, performance was somewhat comparable between model types. ResNet152V2 was the top-performing CNN with an F1-score of 0.6341, while the highest among transformers was Swin (0.6269), closely followed by CvT (0.6076). These results indicate that both CNNs and transformers are capable of identifying neutral expressions with moderate accuracy.

The confusion matrices, as shown in Figure 3, provide further insight into model behavior by showing the distribution of predicted versus actual labels. Correct predictions are found along the diagonal, with misclassifications represented by off-diagonal values.

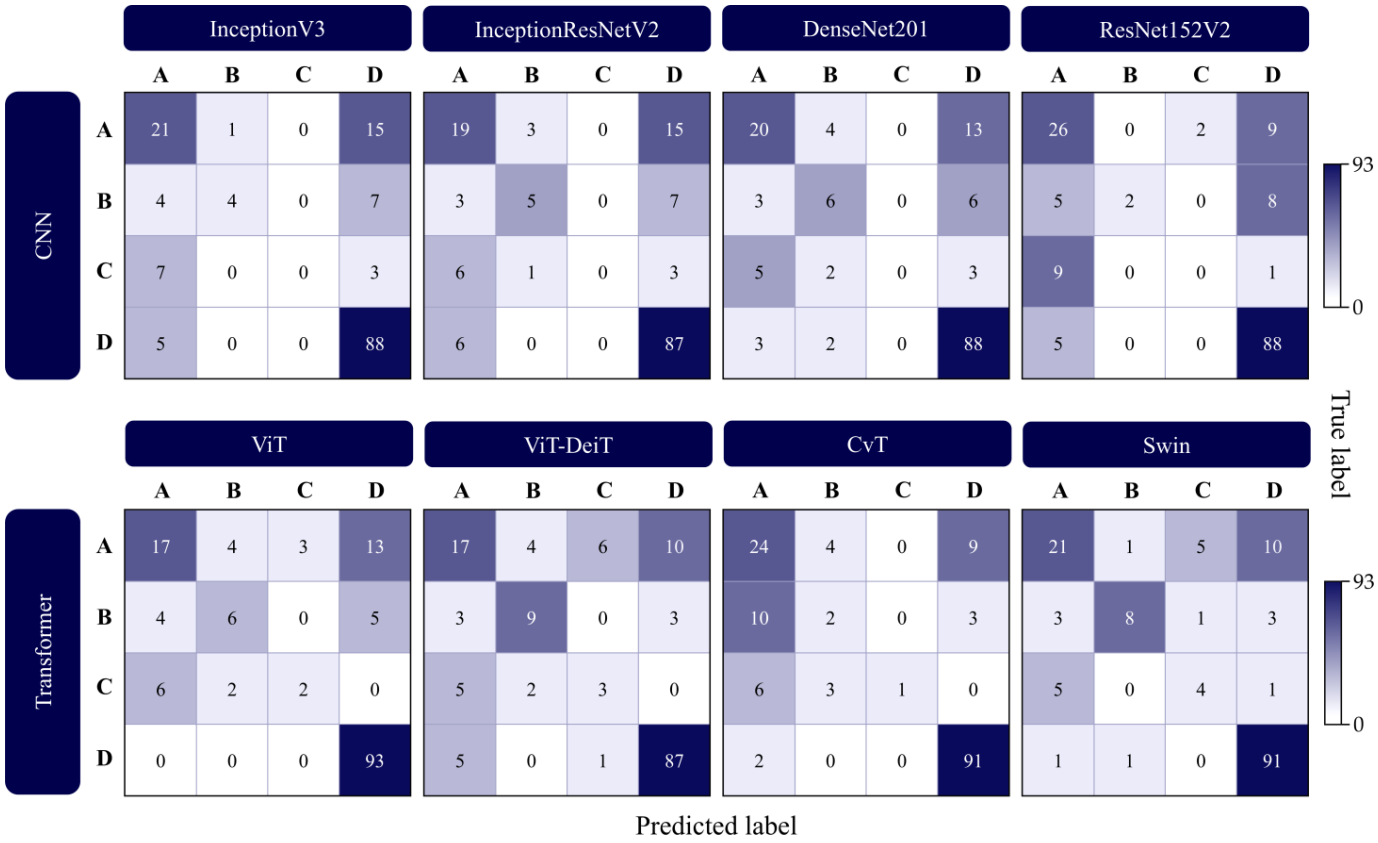

Figure 3. The confusion matrices for the models presented in this study.

Among CNNs, ResNet152V2 showed the most balanced performance, correctly classifying 26 natural, 5 anger, 9 fear, and 88 joy instances. Most CNNs performed well on joy (≥87 correct) but struggled significantly with fear—frequently misclassifying these instances as natural or anger. For example, InceptionV3 misclassified 7 fear instances as natural and failed to predict any correctly.

Transformer models showed marked improvement. The Swin Transformer correctly identified 91 joy, 8 anger, and 1 fear instances, along with 21 natural. Notably, it distributed misclassifications more evenly and with lower frequency, which aligns with its high F1-scores across all emotion categories. ViT-DeiT also performed well in anger recognition (9 correct predictions), while CvT achieved strong natural (24) and joy (91) predictions but misclassified many fear instances. ViT demonstrated strong joy detection with 93 correct predictions and modest improvements in fear and anger compared to CNNs.

A key observation is that all models—even top-performing transformers—continued to struggle with the fear category, reflecting its inherent complexity and possibly lower representation in the dataset. However, transformers displayed greater capacity to generalize across emotions and maintain balanced classification, suggesting they are more suited to the nuanced affective states often exhibited by autistic individuals.

The results clearly demonstrate the superior performance of transformer-based architectures compared to traditional CNN models in recognizing emotions in autistic children. Transformers consistently outperformed CNNs across all metrics, with the Swin Transformer achieving the best overall results. This can be attributed to the inherent architectural differences between transformers and CNNs [41,42]. Transformers utilize self-attention mechanisms that allow them to capture long-range dependencies and global contextual information within images, which is crucial for accurately interpreting subtle and complex facial expressions [43]. In contrast, CNNs rely on localized convolutional filters that focus on spatially limited regions, potentially missing important global cues necessary for nuanced emotion recognition, especially in autistic children whose expressions may be atypical or less exaggerated [44]. Among CNNs, ResNet152V2 performed best, likely due to its deeper architecture and residual connections that enable better feature extraction and gradient flow [45,46]. However, it still lagged behind transformers, suggesting that simply increasing depth may not fully compensate for CNNs' limited ability to model complex relational features across the entire face. Emotion-specific performance differences further elucidate why transformers have an advantage. Joy was the easiest emotion to detect because joyful expressions typically involve distinct, strong visual cues—such as smiling—that are easier to learn and recognize [17]. Transformers' capacity to integrate diverse facial regions likely enhances their ability to capture these consistent features robustly [47]. Fear was the most challenging emotion across all models. Fear expressions can be subtle, often involving micro-expressions or slight muscle tensions that vary widely between individuals, especially among autistic children who may express fear differently or less overtly [37,44]. Transformers showed some capability here due to their holistic attention approach, which can aggregate weaker, dispersed signals better than CNNs' local filters. Nevertheless, the limited data on fear expressions and their inherent subtlety constrained overall detection performance. Anger detection fell between joy and fear in difficulty. Angry expressions can vary significantly in intensity and presentation, which likely requires models to generalize across diverse facial cues—a task that transformers are better equipped to handle given their global context modeling [41]. The confusion matrices confirm that misclassifications predominantly involve fear and anger being mistaken for neutral or other emotions. This suggests that these emotions either share overlapping visual features or are underrepresented in the training data, emphasizing the need for richer datasets.

An important consideration in interpreting these results is the dataset composition and its similarity to previous studies on autistic emotion recognition. In this work, the dataset comprised 770 total images, split into 615 training and 155 validation samples: natural (37), anger (15), fear (10), and joy (93). This mirrors the FERAC study by Afrin et al. [33], which also used 770 images in four classes but with a 90:10 train–validation split, leaving only 5 test samples for fear and 8 for anger. In the study by Talaat et al. [28], the class imbalance was even more pronounced, with some categories—such as anger and fear—having only 3 test samples each in a 758/72 train–validation configuration across six classes. While Afrin et al. [33] applied targeted data augmentation (rotations, flips, shifts, zoom, brightness adjustments) to partially balance the training data, Talaat et al. [28] relied on model robustness without explicit balancing. In all three cases, imbalance persisted in the evaluation set, meaning that aggregate performance metrics could mask substantial performance disparities in minority classes. From a performance perspective, our top-performing model, the Swin Transformer, achieved an accuracy of 0.8000, F1-score of 0.7889, precision of 0.7903, and recall of 0.8000. This exceeds the best result in Afrin et al.'s work, where the DenseNet121 + MobileNetV2 hybrid achieved 0.7500 accuracy, 0.6000 precision, 0.5700 recall, and 0.5600 F1-score [33]. In contrast, Talaat et al. reported substantially higher nominal performance, with their Xception model reaching 0.9523 accuracy, 0.9320 sensitivity, 0.9421 specificity, and 0.9134 AUC [28]. However, these results were obtained on a validation set in which several classes, including anger and fear, contained only three samples each, rendering per-class accuracy highly unstable and limiting the reliability of aggregate measures. Across all three studies, dominant classes such as joy and natural consistently achieved the highest recognition rates, whereas minority and visually subtle categories like fear remained challenging. This recurring trend underscores the need for larger, more balanced datasets and for evaluation strategies, such as cross-validation or external dataset testing, that provide a more reliable assessment of model generalizability.

Despite promising results, several limitations remain. The poor recognition of fear likely stems from dataset imbalance and a lack of distinctive features for this emotion, indicating a need for more fear samples or additional data modalities (e.g., audio or physiological signals) to improve detection. The dataset's size and diversity may also restrict the models' generalizability, as autistic children display a wide range of emotional expressions influenced by individual differences not fully captured here. Increasing participant diversity and emotional contexts would strengthen robustness. The dataset's limited scale also prevented the application of computationally intensive validation schemes such as k-fold cross-validation for both CNNs and ViTs; instead, we mitigated this by employing extensive data augmentation and preprocessing. Furthermore, focusing solely on facial expressions overlooks other important emotional cues, especially since autistic children may express emotions differently. Incorporating multimodal data and context could further enhance recognition performance and clinical applicability. Finally, the absence of multiple training runs with varying seeds precluded formal statistical testing; future work will address this through repeated experiments on larger, multi-center datasets.

5. Conclusions and Future Work

This study investigated and compared the performance of CNNs and transformer-based models for emotion recognition in autistic children, a task complicated by atypical and often subtle emotional expressions. The models were evaluated across key performance metrics, including accuracy, F1-score, precision, and recall. Among CNNs, ResNet152V2 emerged as the best performer, achieving an accuracy of 0.7484 and an F1-score of 0.7048. However, all CNN models consistently failed to recognize the fear emotion and showed limited effectiveness in detecting anger. These limitations highlight the difficulty tradi-

tional models face when dealing with nuanced affective cues commonly exhibited by autistic individuals. Transformer-based models significantly outperformed CNNs across all categories. The Swin Transformer achieved the highest overall performance, with an accuracy of 0.8000 and an F1-score of 0.7889. It also delivered improved recognition of challenging emotions such as fear and anger, where CNNs struggled. Confusion matrix analysis confirmed Swin's ability to distribute errors more evenly and correctly classify a broader range of emotional states. This suggests that the global attention mechanisms of transformer models provide an advantage in understanding complex facial patterns.

Despite these promising findings, limitations remain, including class imbalance and reliance solely on visual data. Future research should explore multimodal approaches that integrate facial imagery with additional signals such as speech, body posture, gaze tracking, and physiological data, which may enhance the recognition of subtle emotions in autistic individuals. Another promising avenue is the incorporation of emotion recognition models into virtual reality (VR) environments for therapeutic and educational purposes. VR-based social simulations could provide controlled, immersive contexts for practicing emotional understanding, while automated emotion recognition could offer real-time feedback to participants and therapists. Such integration would require addressing challenges related to data synchronization, latency, and user comfort but holds considerable potential for enhancing intervention effectiveness.

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